Faithful Abstractive Malayalam Summarization: Comparative Evaluation of Transformer Models with Hallucination Analysis

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***Abstract*—Abstractive summarization of low-resource languages poses significant challenges due to a lack of annotated data, inadequate domain adaptation, and possible factual errors in generated summaries. While Malayalam is an extensively used Indian language, it has not received adequate attention in the literature on transformer-based summarization. This paper explores the problem of faithful abstractive Malayalam summarization using various transformer models under different paradigms of inference and training.**

Four transformer models were evaluated under zero-shot prompting, two-shot prompting, supervised fine-tuning, and retrieval-augmented generation, thus providing us with sixteen experiments. Model performances were evaluated using traditional summarization metrics such as ROUGE, BLEU, and BERTScore, in addition to a reliability-aware evaluation framework based on faithfulness, context precision, context recall, hallucination rate, and a novel hallucination score metric.

The results indicate that fine-tuned models generally performed better compared to prompting and retrieval-augmented approaches both in terms of summarization quality and reliability. Among all fine-tuned models, fine-tuned mT5 demonstrated the best balance in terms of performance metrics, achieving the highest faithfulness and the lowest hallucination score, whereas fine-tuned mBART performed the best on lexical overlap measures.

In summary, task-specific model adaptation proved to be superior to prompt engineering and retrieval-augmentation techniques for faithful abstractive summarization in low-resource Malayalam setting.

***Index Terms*—Malayalam Summarization, Abstractive Summarization, Hallucination Detection, Retrieval-Augmented Generation, Transformer Models, Faithfulness Evaluation, Low-Resource NLP**

I. Introduction

Abstractive summarization has emerged as a prominent natural language processing task due to the rising volume of digitized textual data. Unlike extractive summarization, which involves selecting relevant sentences from the source text, abstractive summarization generates shorter and more coherent summaries through paraphrasing, producing human-like results. However, even with the development of transformer-based language models, abstractive summarization of low-resource languages continues to face several challenges.

Malayalam, a highly inflected Dravidian language, has received relatively little attention in the context of abstractive summarization research owing to a lack of benchmark datasets and minimal task-specific adaptations of multilingual transformer models. Although architectures like mBART, mT5, BLOOMZ, and Sarvam have shown impressive multilingual generation performance, their suitability for Malayalam summarization greatly depends on the chosen adaptation method.

One of the most critical problems associated with abstractive summarization is that of hallucinations, when summaries produced by transformers contain unsupported or even false statements even though they appear grammatically correct. Traditional evaluation methods like ROUGE and BLEU focus only on lexical similarity and thus cannot reliably detect factual inconsistencies, emphasizing the need for hallucination-aware evaluation.

In this paper, we conduct an analysis of transformer-based abstractive summarization of Malayalam in four experimental settings: zero-shot prompting, two-shot prompting, supervised fine-tuning, and retrieval-augmented generation (RAG). Sixteen model configurations were compared using traditional summarization metrics and factual consistency metrics. In addition, a special hallucination score was designed to estimate unsupported generation and investigate the summary quality-faithfulness tradeoff.

The key contributions of this paper are:

- Comparative evaluation of sixteen transformer-based Malayalam summarization configurations.
- Analysis of zero-shot, two-shot, fine-tuned, and retrieval-augmented summarization strategies.
- Hallucination-aware evaluation using faithfulness, context precision, context recall, and semantic similarity metrics.
- Introduction of a custom hallucination score for factual reliability assessment.

- Benchmark insights for trustworthy abstractive summarization in low-resource Indic languages.

The remainder of this paper is organized as follows. Section II reviews related work in Malayalam summarization and hallucination analysis. Section III describes the dataset, preprocessing, and experimental methodology. Section IV presents the evaluation metrics used for quality and factual consistency assessment. Section V discusses experimental results and comparative analysis. Section VI concludes the work and outlines future research directions.

II. Related Work

The process of text summarization has seen significant development over time from extractive methods to advanced abstractive approaches that utilize deep learning. The first study on abstractive summarization was presented by Gangadharan et al. [7], who focused on graph-based conceptual mining techniques. The second paper, written by Chandu et al. [8], was dedicated to the problem of extractive query-based summarization.

Regarding Indian languages, Malayalam summarization is still an unsolved problem caused by morphological complexity and lack of benchmark datasets. Nambiar and Sindhya [1] introduced an attention-based encoder-decoder approach to abstractive summarization for Malayalam. Later, the same problem was solved by adding part-of-speech tagging information into the attention mechanism framework, thus improving the quality of the results obtained [2].

In order to solve the problem with the lack of standardized datasets, Rahul Raj and Dhanya [3] developed the Social-sum-Mal dataset, which could be used for benchmarking in the area of Malayalam abstractive summarization.

Current trends in transformer-based text summarization are related to pretrained and large language models. Fine-tuning transformer models for abstractive summarization has shown excellent results for all Indian languages [10]. Moreover, the applicability of LLaMA-based summarization was proved by Tanev et al. [5].

Abstractive summarization is not only used in Malayalam language but also can be applied in specific domains, including legal documents [6] and medical documents [9]. Parikh et al. [4] conducted a multilingual review of transformer-based summarization, emphasizing the dominance of these architectures and highlighting open problems, such as factual consistency, semantic preservation, and evaluation.

Hallucination, faithfulness, and factual reliability remain poorly investigated in the field of Malayalam abstractive summarization when comparing different methods, such as fine-tuning, zero-shot prompting, and retrieval-augmented summarization.

III. Methodology

A. Dataset

The dataset used in this study was constructed by combining two Malayalam abstractive summarization resources: the **Social-Sum-Mal** dataset [3] and the **AI4Bharat Malayalam Sentence Summarization dataset**.

The Social-Sum-Mal dataset originally contained multiple metadata and summarization-related fields, including `chapter_no`, `para_no`, `heading`, `input`, `long_summary`, `extreme_summary`, `query`, `answer_summary`, and `topic`. For this work, only the `input` field (source article text) and the `long_summary` field (reference abstractive summary) were selected, as they were most suitable for full-length abstractive summarization.

The AI4Bharat dataset provided Malayalam article-summary pairs in a different schema. To ensure consistency, both datasets were standardized into a unified article-summary format, where source article text was treated as input and the corresponding summaries were used as reference outputs.

After merging both datasets, the combined corpus contained over **16,000 article-summary pairs**. A preprocessing pipeline was applied to address dataset heterogeneity, including duplicate removal, missing value filtering, malformed text cleaning, mixed-language sample removal, and elimination of low-information records. After cleaning, the final dataset consisted of **9,698 high-quality Malayalam article-summary pairs**.

The cleaned dataset was randomly split into training, validation, and testing partitions using an 80:10:10 ratio for consistent experimentation.

TABLE I
Dataset Distribution

Split	Samples
Training	7758
Validation	970
Testing	970
Total	9698

B. Data Preprocessing

The data was processed through a structured preprocessing pipeline to enhance its quality prior to conducting experiments. It involved schema standardization of both datasets, elimination of duplicates and missing data, normalization of text, exclusion of multilingual summaries with English words, and exclusion of very short summaries. Subsequently, the data was shuffled and divided based on a predetermined random seed.

This resulted in the development of a reliable Malayalam corpus for prompt-based testing, supervised fine-tuning, and retrieval-augmented training. The biggest problem faced during the creation of the dataset was the heterogeneity of the source datasets.

C. Experimental Configurations

Four multilingual models were evaluated under four experimental settings, producing sixteen total configurations.

D. Experimental Pipeline

The experimental workflow followed a comparative evaluation framework for Malayalam abstractive summarization

TABLE II
EXPERIMENTAL MODEL GROUPS

Group	Approach	Models
Zero-shot	Prompt-only inference	BLOOMZ, mT5, mBART, Sarvam
Two-shot	Example-guided prompting	BLOOMZ, mT5, mBART, Sarvam
Fine-tuned	Supervised adaptation	BLOOMZ, mT5, mBART, Sarvam
RAG	Retrieval-grounded generation	BLOOMZ, mT5, mBART, Sarvam

across multiple transformer architectures and inference strategies.

The pipeline consisted of the following stages:

1) **Dataset Preparation:** Social-Sum-Mal and AI4Bharat Malayalam summarization datasets were merged, cleaned, and split into training, validation, and testing sets.
2) **Model Selection:** These models were selected to represent diverse transformer paradigms, including encoder-decoder multilingual summarization models (mBART, mT5), instruction-following generative models (BLOOMZ, Sarvam), and architectures with differing adaptation behaviour under low-resource Malayalam summarization. Four transformer models were selected for experimentation:
   - mBART
   - mT5
   - BLOOMZ
   - Sarvam
3) **Summarization Strategies:** Each model was evaluated under four experimental settings:
   - Zero-shot prompting
   - Two-shot prompting
   - Supervised fine-tuning
   - Retrieval-Augmented Generation (RAG)
4) **Summary Generation:** Summaries were generated for the test set under each configuration, resulting in 16 model-setting combinations.
5) **Evaluation:** Generated summaries were evaluated using lexical, semantic, and hallucination-focused metrics including ROUGE, BLEU, BERTScore, faithfulness, context precision, context recall, hallucination rate, and custom hallucination score.
6) **Comparative Analysis:** Final results were compared across all configurations to analyze summarization quality and factual consistency.

The workflow consists of dataset preprocessing, execution under four experimental paradigms, summary generation, and comparative evaluation.

E. Fine-Tuning Framework

Fine-tuned models were trained using supervised sequence-to-sequence learning on the prepared Malayalam article-summary corpus. For encoder-decoder architectures (mBART and mT5), source articles were tokenized as model inputs and reference summaries were used as target sequences. Training involved tokenizer-based preprocessing, input truncation,

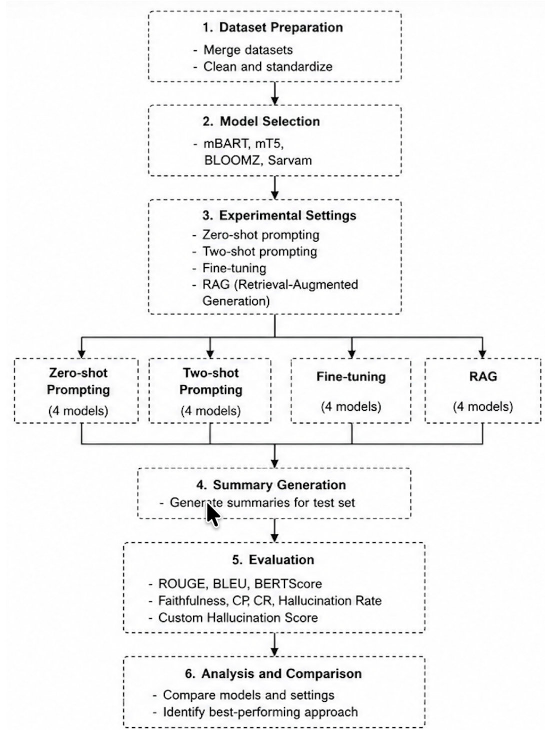

Fig. 1. Overall experimental pipeline for comparative Malayalam abstractive summarization evaluation.

batch-wise optimization, and periodic validation to monitor performance. Model checkpoints were saved based on validation performance to retain the best-performing configuration.

For instruction-tuned causal language models such as BLOOMZ and Sarvam, supervised fine-tuning was performed using prompt-formatted article-summary pairs to align generation with the summarization task. This fine-tuning stage enabled better adaptation to Malayalam summarization patterns, improved contextual understanding, and reduced unsupported content generation compared to zero-shot and prompt-based inference.

F. Fine-tuned Model Training Analysis

To analyze convergence behaviour during supervised adaptation, training and validation loss curves were monitored for all four fine-tuned transformer models. These plots provide insight into optimization stability, learning progression, and potential overfitting behaviour.

As shown in Fig. 6, mT5 and Sarvam exhibited stable convergence with gradually decreasing validation loss, indicating effective learning and reasonable generalization. Fine-tuned mBART demonstrated consistent reduction in training loss with stable validation behaviour, reflecting successful adaptation to Malayalam summarization. BLOOMZ initially showed decreasing loss but later exhibited rising validation loss, suggesting mild overfitting during later training stages.

Overall, the loss curves confirm that supervised fine-tuning significantly improved task adaptation compared with zero-shot and prompt-based settings, leading to stronger summarization performance and improved factual consistency.

G. Retrieval-Augmented Generation

The Retrieval-Augmented Generation (RAG) technique was used in the current work in order to increase factual grounding during summarization. In contrast to the usage of the internal knowledge of the pretrained model, contextual chunks were retrieved from the source article based on semantic similarity search. The source article was broken down into several text chunks that were embedded into vector space and indexed in order to be quickly searched for by their semantic similarity.

The selected chunks were added to the query in order to provide evidence to the model while generating summaries. Thus, the process of generation could be grounded in the retrieved evidence instead of implicit memorization of the information. It was expected that this would help to avoid factual inconsistencies and increase the contextuality of the generated texts.

IV. Evaluation Metrics

A. Summary Quality Metrics

Metrics used to assess summary quality are the standard automatic summarization metrics widely used in the field of abstractive summarization. While there is no single metric that can measure all the aspects of summary quality, a combination of lexical overlap and semantic similarity metrics was chosen to conduct the assessment.

- **ROUGE-1:** Measures the unigram overlap of the generated summary and the reference summary, showing general summary content coverage.
- **ROUGE-2:** Measures bigram overlap, indicating summary phrase-level coherence and structure.
- **ROUGE-L:** Evaluates the longest common subsequence of the generated and reference summaries, showing the structural and sentence-level similarities.
- **BLEU:** Measures the n-gram precision overlap, providing another angle on generation accuracy.
- **BERTScore:** Measures semantic similarity using contextual transformer embeddings, allowing for better evaluation of summary content outside of strict lexical matches.

ROUGE metrics were chosen due to their wide popularity as benchmark measures in summarization tasks, which make it possible to assess both content overlap and structure similarities. BLEU was used to complement ROUGE metrics by providing another n-gram measure of generation accuracy, while BERTScore was used to overcome the limitation of purely lexical metrics and assess summary alignment.

B. Hallucination Evaluation Metrics

To evaluate factual consistency in addition to standard lexical overlap metrics, a halluncination-aware evaluation methodology was applied. The following metrics were used:

- **Faithfulness (F):** Measures semantic consistency of the generated summary with the source article and, therefore, its factual consistency.

For RAG experiments, the context precision and context recall metrics were calculated using the retrieved contextual passages. However, for zero-shot, two-shot, and fine-tuned variants of the model, which do not involve an explicit retrieval of contextual information, these metrics were approximated by measuring semantic similarity between the generated summary and the original source article.

To unify the hallucination evaluation by having a single numerical measure of hallucination severity, the hallucination score was introduced as:

$$HS = 1 - (0.4F + 0.2CP + 0.2CR + 0.2B) \qquad (1)$$

where:

- F = Faithfulness
- CP = Context Precision
- CR = Context Recall
- B = Normalized BERTScore

The lower the hallucination score, the higher factual consistency is.

V. Results and Discussion

A. Overall Performance Comparison

Table III presents the comparative evaluation of all sixteen Malayalam summarization configurations across zero-shot prompting, two-shot prompting, supervised fine-tuning, and pretrained retrieval-augmented generation (RAG) settings.

Performance differences are evident based on experimental setup. Fine-tuned models performed the best in terms of both summarization quality and factual consistency, with mBART, mT5, and BLOOMZ achieving the best summarization scores. Fine-tuning facilitated direct adaptation to the language summarization style, leading to enhanced source-summary correlation and grounded generation.

Among fine-tuned models, mT5 achieved the lowest hallucination score, suggesting the best trade-off between summary quality and factual consistency, while mBART had the highest lexical overlap scores. BLOOMZ also had relatively good performance after adaptation, but Sarvam had poorer performance, which could be attributed to its architecture and unsuitability for encoder-decoder summarization.

Zero-shot and two-shot prompting led to low ROUGE and BLEU scores since pretrained multilingual models were not trained specifically for Malayalam summarization. Paraphrasing was observed in generated summaries that preserved semantic meanings but were lexically different from the references.

Two-shot prompting helped in improving performance in some cases, especially mT5, where structure-based generation was induced using example-based prompting. However, the improvement in performance was still limited when compared with supervised learning.

RAG-based models resulted in mixed results. Although context retrieval helped in grounding the generated summary, the performance was still behind that of fine-tuned models since retrieval provides external evidence to assist the summarization

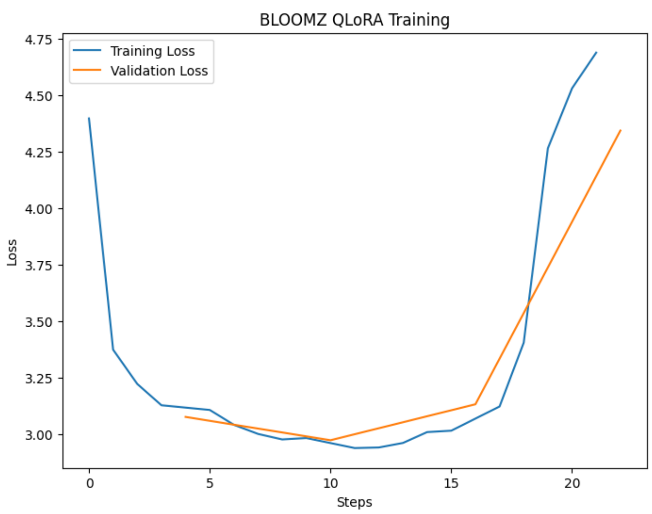

Fig. 2. (a) BLOOMZ Fine-tuning

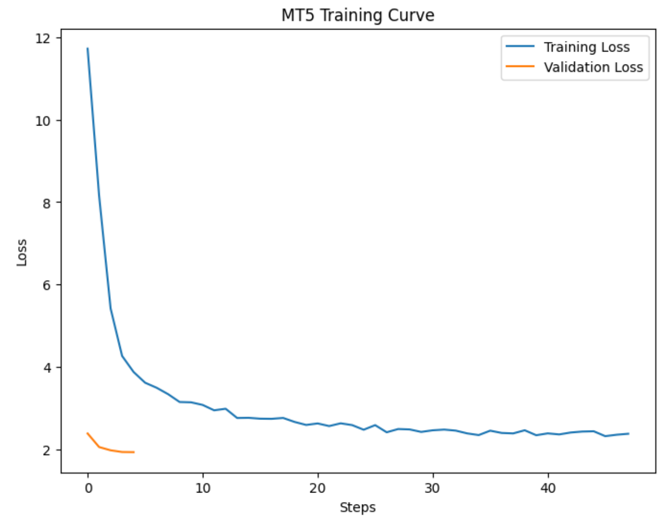

Fig. 3. (b) mT5 Fine-tuning

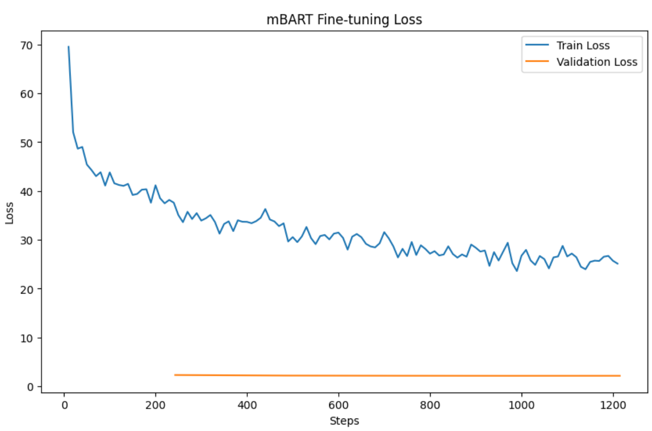

Fig. 4. (c) mBART Fine-tuning

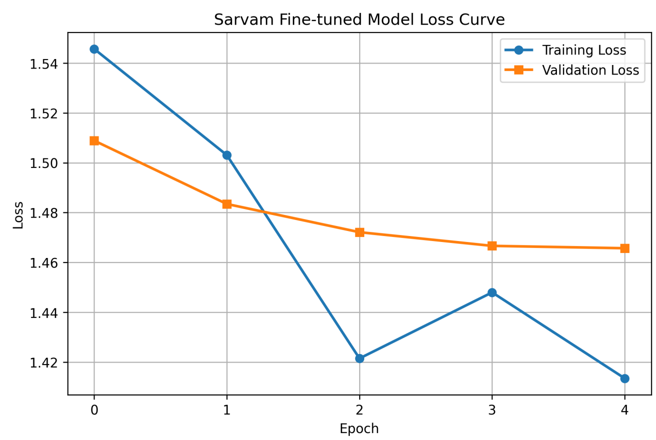

Fig. 5. (d) Sarvam Fine-tuning

Fig. 6. Training and validation loss curves for fine-tuned transformer models used in Malayalam abstractive summarization.

TABLE III
OVERALL COMPARATIVE PERFORMANCE ACROSS ALL SUMMARIZATION CONFIGURATIONS

Model	Setting	R1	R2	RL	BLEU	BERT	Faith	CP	CR	HS	HR
Sarvam	Zero-shot	0.0022	0.0001	0.0022	0.8024	0.8161	0.8549	0.9670	1.0000	0.1014	0.0000
BLOOMZ	Zero-shot	0.0137	0.0000	0.0137	1.0683	0.7958	0.8116	0.9928	1.0000	0.1176	0.0000
mT5	Zero-shot	0.0003	0.0000	0.0003	0.0251	0.7682	0.7636	1.0000	1.0000	0.1409	0.0000
mBART	Zero-shot	0.0143	0.0017	0.0145	0.6864	0.8097	0.8478	0.8990	1.0000	0.1191	0.0000
Sarvam	Two-shot	0.0005	0.0000	0.0005	0.3420	0.8170	0.8388	1.0000	1.0000	0.1011	0.0000
BLOOMZ	Two-shot	0.0100	0.0000	0.0100	0.0240	0.7738	0.7724	0.9918	1.0000	0.1379	0.0000
mT5	Two-shot	0.0044	0.0004	0.0044	0.9579	0.8349	0.8747	1.0000	1.0000	0.0831	0.0000
mBART	Two-shot	0.0065	0.0000	0.0062	1.5248	0.8023	0.8187	0.8247	1.0000	0.1471	0.0000
mBART	Fine-tuned	0.0377	0.0017	0.0375	14.4269	0.8852	0.9122	1.0000	1.0000	0.0581	0.0000
mT5	Fine-tuned	0.0325	0.0010	0.0329	14.1766	0.8799	0.9222	0.9990	1.0000	0.0553	0.0000
BLOOMZ	Fine-tuned	0.0378	0.0039	0.0380	7.6232	0.8680	0.9045	0.9990	1.0000	0.0648	0.0000
Sarvam	Fine-tuned	0.0085	0.0002	0.0084	4.7570	0.8455	0.8546	0.9959	1.0000	0.0899	0.0000
BLOOMZ	Pretrained+RAG	0.0078	0.0000	0.0077	0.2189	0.7799	0.7978	0.9969	1.0000	0.1255	0.0000
mT5	Pretrained+RAG	0.0011	0.0000	0.0010	0.3726	0.7825	0.7870	1.0000	1.0000	0.1287	0.0000
mBART	Pretrained+RAG	0.0067	0.0000	0.0067	0.1644	0.7900	0.8229	0.6990	1.0000	0.1730	0.0000
Sarvam	Pretrained+RAG	0.0076	0.0003	0.0076	3.6735	0.5195	0.5864	0.5080	0.5080	0.4583	0.6969

process without changing the intrinsic capability of the model. The worst-performing RAG model was Sarvam, with poor semantic consistency and the only hallucination rate among the tested models.

B. Group-wise Analysis

Zero-shot prompting yielded baseline performance of multilingual summarization without any adaptation to the task, thus, resulting in lower lexical overlap and factual consistency. On the other hand, two-shot prompting helped improve generation quality in some models through example-based prompting.

Fine-tuned models performed the best in all tasks since they were directly adapted using supervised learning from Malayalam article-summary pairs. In addition, RAG-based models helped in improving contextual grounding, although the extent of improvement depended on model compatibility.

The near-zero hallucination rates of most configurations are attributed to the conservative binary threshold used in the hallucination detection process, where only extreme cases of inconsistency were categorized as hallucinations. Thus, the hallucination score provides a finer evaluation of factual consistency.

C. Comparison of Hallucination Scores in Prompt-based Non-RAG Models

To further evaluate factual consistency in prompt-based summarization, the hallucination scores of zero-shot and two-shot non-RAG models are compared in Fig. 7.

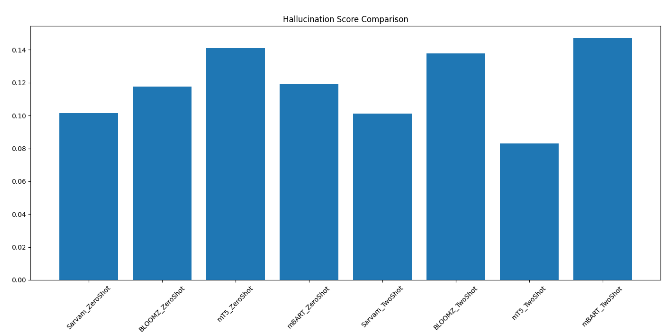

Fig. 7. Hallucination score comparison across zero-shot and two-shot non-RAG transformer models. Lower values indicate better factual consistency.

It is apparent that there is some variation in the architecture of the summarization models used. Models using two-shot prompting technique had the lowest hallucination scores, indicating their greater factual grounding capabilities, compared to those using two-shot prompting and zero-shot prompting techniques.

D. Normalized Comparative Metric Heatmap

To give a comprehensive comparison of all sixteen models, a heatmap is constructed based on all the metrics.

Since the evaluation metrics have different scales, directly visualizing using raw metric scores would be misleading. Instead, min-max normalization was done for each individual metric separately for color scaling purposes, while the actual metric value is preserved as annotations.

Factual consistency was defined as follows:

$$\mathrm{FC} = 1 - \mathrm{HS} \tag{2}$$

where HS is the hallucination score.

The heatmap shown in Fig. 8 clearly illustrates the good performance of fine-tuned encoder-decoder models, such as mT5 and mBART, in both the quality and factual consistency domains.

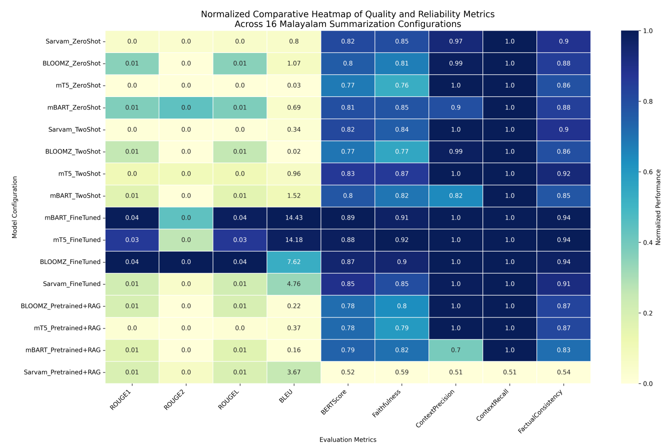

Fig. 8. Normalized comparative heatmap of summarization quality and factual reliability metrics across all configurations. Higher values indicate better performance.

E. Best Model Discussion

In all of the configurations, fine-tuned **mT5** performed the most consistently in terms of summarization quality and factual reliability. While fine-tuned mBART had slightly higher ROUGE scores, fine-tuned mT5 had the best faithfulness score (**0.9222**) and the lowest hallucination score (**0.0553**), implying high factual consistency.

This can be attributed to the encoder-decoder text-to-text architecture of mT5, which suits abstractive summarization well. Fine-tuning also enhances the source-document summary alignment and Malayalam adaptation, thus allowing for more grounded generation.

Overall, in light of both summarization quality and factual consistency, fine-tuned **mT5** performed the best out of all configurations.

F. Error Analysis

One of the challenges noted was the consistently poor ROUGE scores in prompt-based and RAG setups. This is because of the abstractive nature of Malayalam summarization, whereby paraphrasing and morphological variation lead to low lexical overlap despite the retention of semantic meaning.

Prompt-based models sometimes produced fluent yet poorly aligned summaries without task-specific adaptation. RAG models also performed inconsistently when the retrieved context is only partially relevant. The poor performance of Sarvam in the RAG setup implies ineffective fusion of evidence retrieval and summarization tasks.

These suggest that the model architecture, retrieval quality, and task adaptation strategies are crucial factors influencing summarization reliability in the Malayalam environment.

VI. Conclusion

In this paper, **comparative analysis of sixteen Malayalam abstractive summarization configurations** comprising zero-shot prompting, two-shot prompting, supervised fine-tuning, and retrieval-augmented generation was conducted using four transformer architectures: mBART, mT5, BLOOMZ, and Sarvam. Factual consistency evaluation was incorporated to evaluate the summarization quality and reliability for Malayalam.

Experimental results revealed significant performance difference among various paradigms. Prompt-based zero-shot and two-shot models were found to exhibit reasonable semantic generation capabilities, although they were less accurate in lexical alignment and factual consistency. Retrieval-augmented generation improved the grounding capabilities of some of the pretrained models, but the improvement was inconsistent across different architectures and relied heavily on the quality of retrieval and model-task alignment. In contrast, fine-tuned models consistently outperformed others, with their Malayalam adaptation leading to more consistent performance.

Among all of the summarization configurations, **fine-tuned mT5 exhibited the most consistently good performance**, with the highest faithfulness score and lowest hallucination score, while fine-tuned mBART having the highest ROUGE scores. This shows that supervised fine-tuning is significantly more effective than prompt-based adaptation or retrieval-augmentation in generating more faithful abstractive summaries in the Malayalam environment.

The hallucination-aware evaluation framework adopted in this paper also allowed for deeper analysis beyond the common ROUGE scores through the explicit measurement of factual grounding and unsupported generation.

VII. Future Work

Future work will focus on larger Malayalam benchmark datasets, stronger retrieval mechanisms, improved hallucination detection with human evaluation, and domain-specific adaptation for more reliable summarization. Retrieval-aware fine-tuning and larger multilingual instruction-tuned models may further improve factual consistency and summary quality.

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